Natural-text filter calibration: ruling out filter pathology in F2.1

May 13, 2026

1 Question

The F2.1 figure in `results/counter_decode/figures/F2.1_calibration_gap.png` shows the streaming Laplace $\hat{\beta}$ filter drifting substantially during β-decoded generation: $\log \hat{\beta}$ reaches $\approx +1.3$ at $\beta_{\mathrm{dec}} = 2$ and ≈ -3.3 at $\beta_{\mathrm{dec}} = 0.5$ by $t = 199$, growing with t and anti-symmetric in $\log \beta_{\mathrm{dec}}$. The framework reads this as evidence the model's response to β-decoded prefixes is non-multiplicative. A weaker alternative: the filter has baseline drift even with no temperature manipulation, and the F2.1 gap is partly the filter's own pathology rather than a property of the imprint. This experiment rules out the weaker alternative.

2 Setup

Same model (Qwen-2.5-3B), same filter parameters ($\sigma_0 = 0.5$, log-normal Laplace, $\alpha = \gamma = 1$) as F0/F2. Four conditions, 64 passages each, 200 filter updates per passage:

- **WikiText / self_gen** — prompt = first 20 tokens of a WikiText-103 article; the model samples the next 200 tokens at $\beta_{\mathrm{dec}} = 1$; filter sees (ℓ_t, x_t) with $x_t \sim \mathrm{softmax}(\ell_t)$.
- **WikiText / natural** — same 20-token prompt; the next 200 tokens are the real article continuation.
- **WritingPrompts / self_gen** — same protocol as WikiText/self_gen but with WritingPrompts stories as the prompt source.
- **WritingPrompts / natural** — same as WritingPrompts/self_gen but x_t is the real story token.

The two *self_gen* arms are the actual no-imprint null consistent with F2.1's sampling protocol — the only way to make x_t i.i.d. with respect to $\mathrm{softmax}(\ell_t)$, which is the distribution under which the filter's score $\ell_t[x_t] - E_q[\ell_t]$ is mean-zero by construction. The two *natural* arms test the model's calibration on out-of-distribution text.

The headline x-axis is cumulative Fisher $\Sigma_\tau \mathrm{Var}_q(\tau) = J_t - J_0$ (the filter's effective sample size), not t. Different corpora accumulate J at different rates because of entropy differences; using ΣFisher controls for that. Each condition is shown over its own data range so we don't hide tail behavior of low-entropy corpora.

3 Headline result

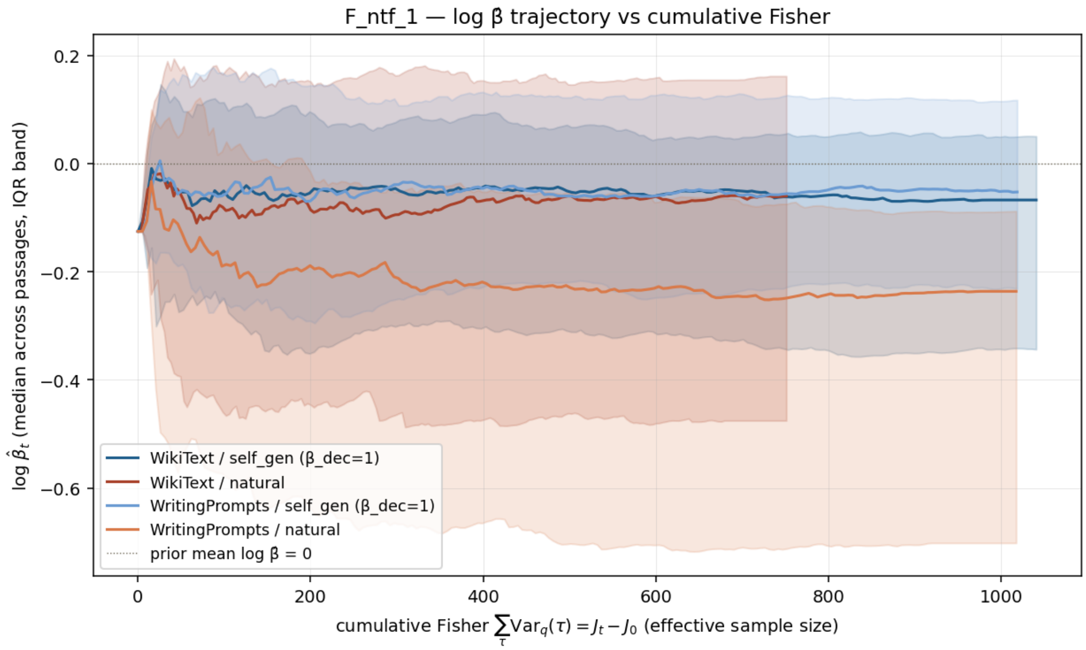

Figure 1: Median $\log \hat{\beta}_t$ across passages, with 25/75% IQR band, plotted against cumulative Fisher. Self_gen arms (blue tones) flatline near zero across both corpora; WikiText/natural tracks self_gen tightly; WritingPrompts/natural drifts visibly negative and stays there.

Table 1: Asymptotic $\log \hat{\beta}$ per condition (median, IQR across passages, SE of median, and trimmed median dropping the 3 worst tails on each side).

condition	N	median	IQR	SE(med)	trimmed med
WikiText / self_gen	64	-0.070	[−0.334, +0.052]	0.068	-0.070
WikiText / natural	64	-0.065	[−0.475, +0.162]	0.093	-0.065
WritingPrompts / self_gen	64	-0.052	[−0.255, +0.127]	0.054	-0.052
WritingPrompts / natural	64	-0.236	[−0.702, −0.089]	0.074	-0.236

Three structural observations.

(a) Self_gen flatlines on both corpora. Median $\log \hat{\beta}_\infty = -0.070$ (WikiText/self_gen, $t = -1.03$ vs. zero) and -0.052 (WritingPrompts/self_gen, $t = -0.96$). Both are within 1 SE of zero. This says: under self-sampling at $\beta_{\text{dec}} = 1$ — the no-imprint null — the filter shows no baseline drift detectable at this sample size. The data is consistent with the filter being asymptotically unbiased.

(b) WikiText/natural agrees with self_gen. Paired difference WikiText/natural − WikiText/self_gen $= -0.050$ (mean), SE $= 0.096$, $t = -0.52$. The two arms are statistically indistinguishable on Wikipedia. Qwen-2.5-3B is well-calibrated there: real Wikipedia next-tokens look exactly as the model expects under $\mathrm{softmax}(\ell_t)$.

(c) WritingPrompts/natural drifts negative, robustly. Median $\log \hat{\beta}_\infty = -0.236$ on the natural arm. Paired against WritingPrompts/self_gen — holding the prompt fixed and varying only the source of x_t — the difference is -0.304 (mean), SE $= 0.080$, $t = -3.80$ ($p < 10^{-3}$). The trimmed median (dropping 3 outliers per tail) equals -0.236, identical to the untrimmed median, so the offset isn't driven by a few catastrophic passages. Qwen-2.5-3B is over-confident on actual Reddit creative writing: real WritingPrompts tokens systematically have lower logits than softmax(ℓ_t) would predict.

4 Secondary diagnostics

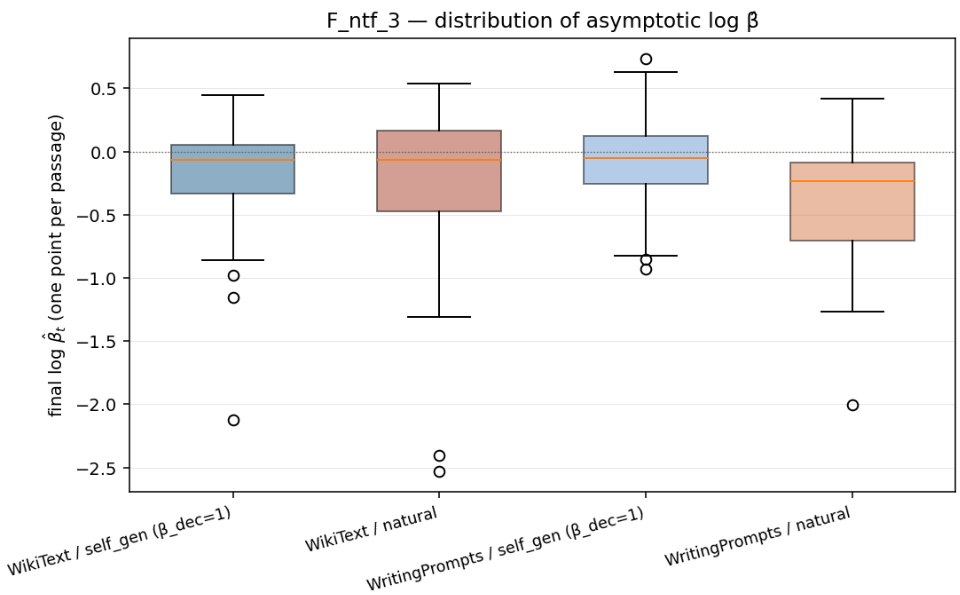

Figure 2: Per-passage $\log \hat{\beta}_\infty$ across the four conditions. The WritingPrompts/natural box sits visibly below zero; the others straddle it.

5 What this implies for F2.1

This experiment establishes one narrow but useful fact: *the filter has no detectable baseline drift on i.i.d. samples from the model's own distribution*, on either corpus. F2.1 shows $|\log \hat{\beta}|$ reaching 1.3–3.3 at extreme β_{dec} values, which is at least an order of magnitude larger than the largest natural-text offset we measured (-0.236 on WritingPrompts). So *the F2.1 drift is not filter-side baseline pathology.*

This is strictly narrower than "the F2.1 drift is imprint structure." The natural-text experiment does not test what the filter does when fed β-decoded prefixes that may be non-multiplicatively distorted relative to the imprint model. A filter that correctly handles self-sampled data can still misattribute a non-multiplicative distortion to β if the imprint hypothesis is wrong. So F2.1's drift could reflect (i) a real multiplicative imprint that the filter recovers correctly, or (ii) a non-multiplicative distortion that the filter interprets through its imprint prior. This experiment cannot distinguish (i) from (ii). It only rules out (iii) baseline filter drift unrelated to the decoder.

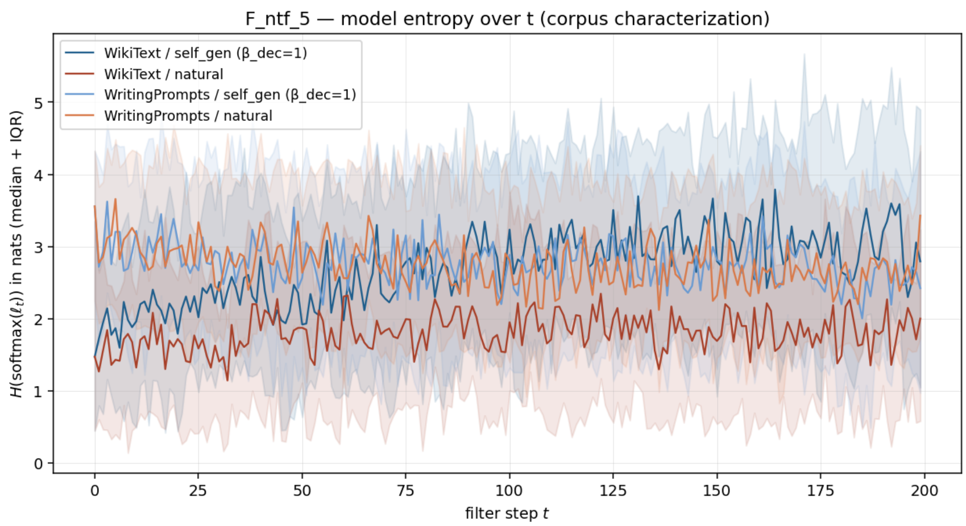

Figure 3: Model entropy $H(\text{softmax}(\ell_t))$ per step, median + IQR. WikiText/natural has the lowest entropy (1.983 nats), confirming that real Wikipedia is more predictable to the model than its own self-generated continuations from the same prompts.

6 A separate finding: out-of-distribution calibration offset

Independently of the F2.1 question, the natural arm establishes that Qwen-2.5-3B has a small but real over-confidence offset on Reddit creative writing (-0.304 in log-β space against the self-gen baseline, paired $t = -3.80$) and no detectable offset on Wikipedia (paired $t = -0.52$). This is a confound for any future use of the filter on out-of-distribution corpora.

Whether this offset contaminates the F2.1 result on the existing F0 prompts is *not measured here*. The F0 prompts (`src/decoding_decoding/prompts.py`) include fiction and dialogue genres that may inherit the WritingPrompts-like offset. Verifying this would require running the natural-text filter on held-out natural continuations of each F0 prompt — a small follow-up experiment.

7 Limitations

- **Two corpora is two corpora.** The headline claim "filter is well-specified under self-sampling" is corpus-independent, but the magnitude of the WritingPrompts offset reflects that one dataset.
- **Trajectory-to-trajectory variance is wide.** Per-passage IQRs span ~ 0.4 in log-β space, dominated by the first few filter steps where J_t is small and individual updates move $\hat{\beta}$ a lot. Median-based statistics are stable, but tail behavior is undersampled at $N = 64$.
- **The natural-text test doesn't reach the framework's load-bearing claim.** F2.1 shows non-multiplicative behavior under β-decoding; this experiment shows the filter doesn't invent that behavior on natural text. It does not isolate whether the β-decoded drift is faithful multiplicative imprint or some other distortion the filter is reading through its prior.

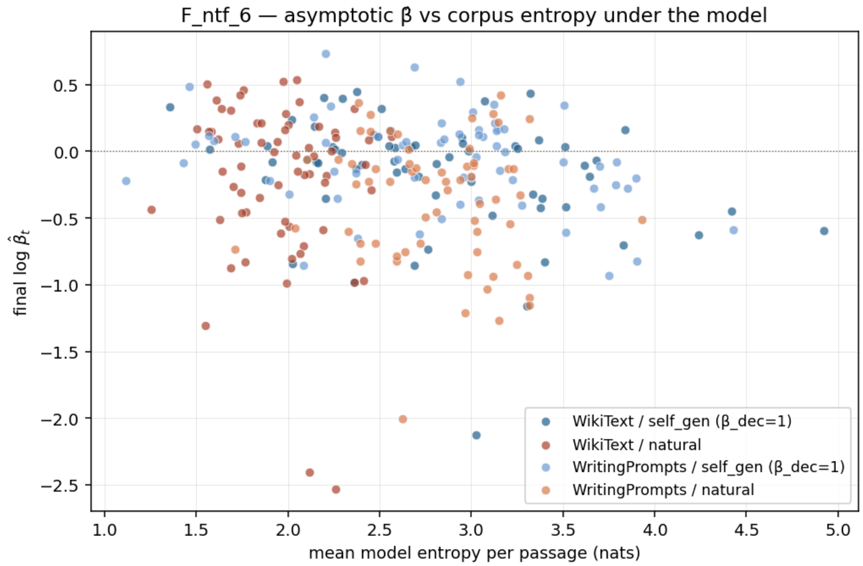

Figure 4: Final log $\hat{\beta}$ vs. mean model entropy per passage. No clean monotone relationship across the cloud; the WritingPrompts/natural cluster sits at moderate entropy with systematically lower log $\hat{\beta}$.

8 Bottom line

What this rules out: the F2.1 calibration gap is not artifact of filter-side baseline drift. On i.i.d. samples from softmax(ℓ_t), the filter holds log $\hat{\beta}$ within ~ 0.07 of zero (within 1 SE) regardless of corpus.

What this rules in: a small (~ 0.3 log-units) calibration offset on creative-writing text that should be treated as a confound on any future filter use on out-of-distribution data.

What this does not address: whether the F2.1 drift is true multiplicative imprint vs. a non-multiplicative distortion the filter is misinterpreting. That requires a different experiment.